Relation-Conditioned Modality Gating for Student Performance Prediction in Heterogeneous Graphs

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Abstract

Predicting student academic performance requires integrating structured records, textual content, and behavioral logs — modalities that carry different signals depending on the relational context in which they are used. Existing heterogeneous graph neural networks (HGNNs) collapse these modalities into a single node embedding and apply uniform message passing regardless of edge type, discarding relation-specific modal relevance. We propose a relation-conditioned modality-gated heterogeneous graph neural network (**MG-HGNN**) in which each node maintains separate modality embeddings and a learnable, per-edge-type softmax gate $\boldsymbol{\alpha}^{(r)}$ controls how much each modality contributes to message passing along relation r. Evaluated on three educational datasets spanning a UK university MOOC (OULAD) and a US mathematics tutoring platform (ASSISTments 2009 and 2015), MG-HGNN consistently outperforms the identical HGTConv backbone without gating across all settings, achieving AUC-ROC 0.8443 ± 0.0292, 0.867 ± 0.019, and 0.950 ± 0.008 respectively. We additionally deploy MG-HGNN (standalone, without re-running baselines) on MOOC-Cube, a large-scale Chinese MOOC platform with 199,199 students and genuine per-course text, reaching AUC-ROC 0.9713 ± 0.0096 across 3 seeds × 5 folds. Gate-weight analysis across datasets shows the mechanism learns non-uniform, relation-specific modality preferences without manual specification; whether it responds to genuinely *informative* text remains an open question, since the text signal available to the gate is a per-student placeholder on all four datasets evaluated, MOOC-Cube included — real text on MOOC-Cube exists only at the course/video level, outside the gate's current student-centric scope.

CCS Concepts

• **Computing methodologies → Neural networks**; *Artificial intelligence*.

Keywords

heterogeneous graph neural networks, student performance prediction, modality gating, educational data mining, learning analytics

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1 Introduction

Student academic performance prediction is a cornerstone task in educational data mining [31] and learning analytics [30], enabling institutions to identify at-risk learners early and deliver timely, targeted interventions. The stakes are high: in large-scale online education, dropout rates frequently exceed 50% [8], and even in campus settings, late detection of struggling students can have lasting consequences on retention and degree completion.

Early approaches to this problem treated students as independent instances, applying classical machine learning methods [25, 26] — logistic regression, support vector machines, and gradient-boosted trees — to flat per-student feature vectors assembled from demographic records, prior grades, and assignment scores. Complementary work in knowledge tracing [22–24] models student mastery at the exercise level using sequential models, but treats each student in isolation without relational graph structure. While effective on structured data, these methods discard the rich relational structure that characterises real educational ecosystems: students interact with courses, collaborate with peers, consume learning resources, and are guided by instructors. Graph neural networks (GNNs) have emerged as the natural framework for capturing these relations, with recent work demonstrating that explicit relational modelling consistently outperforms flat-feature baselines [7, 9, 10].

Despite this progress, a fundamental limitation persists in all existing heterogeneous GNN (HGNN) approaches: they collapse all available data modalities — structured academic records, textual content, and behavioural interaction logs — into a *single, unified node embedding* before message passing begins. This design ignores a key educational insight: different relation types in the student graph are informationally asymmetric. A student's clickstream behaviour on a learning resource is most predictive when propagating information through an *accessed* edge, but the same student's discussion posts and assignment text are more relevant when propagating through a *collaborated-with* edge. Uniform modality fusion across all edge types conflates these distinct signals and limits model expressiveness. Recent work [37] demonstrates the value of jointly modelling spatial and temporal student dependencies; however, homogeneous graph approaches do not capture the typed relational structure — enrolment, access, and collaboration — that distinguishes educational heterogeneous graphs.

This paper proposes MG-HGNN, a *Modality-Gated Heterogeneous Graph Neural Network* that addresses this gap with a single, lightweight architectural addition. Each node maintains three *separate* modality embeddings, produced by a multi-layer perceptron (structured), a frozen BERT encoder (textual), and a bidirectional GRU (behavioural). The core innovation is a per-edge-type *modality gate*: for each relation type $r \in \mathcal{R}$, a dedicated linear layer W_r learns a softmax weight vector $\boldsymbol{\alpha}^{(r)} = [\alpha_s^{(r)}, \alpha_t^{(r)}, \alpha_b^{(r)}]$ that dynamically controls how much each modality contributes to message passing along that specific relation. The gate adds only 4,620 parameters (0.53% of total trainable parameters) and is learned end-to-end with no manual specification of modal preferences.

Contributions. This paper makes four contributions:

(1) We identify and formalise the *modal asymmetry problem* in educational heterogeneous graphs: the empirical observation that different relation types carry different modal signals, which uniform HGNN fusion fails to exploit.
(2) We propose MG-HGNN. To the best of our knowledge, it is the first *educational* HGNN framework that explicitly conditions modality fusion on heterogeneous edge relation types, enabling modality preferences to vary by relational context without manual specification. Unlike concurrent work on multi-modal HGNNs [36], which weights modalities globally per node, MG-HGNN learns a separate softmax gate W_r for each edge relation type r, allowing modal preferences to vary by relational context. This is a strictly finer-grained conditioning than any existing approach.
(3) We evaluate MG-HGNN on three educational datasets — OULAD, ASSISTments 2009, and ASSISTments 2015 — demonstrating consistent improvement over the identical HGTConv backbone without gating across all three, with gate gains of +0.047, +0.009, and +0.014 AUC. A cross-dataset gate-weight analysis reveals the mechanism adapts to actual modality informativeness, upweighting behavioral features on OULAD without any manual specification. On ASSISTments, where text is similarly absent, gate weights show reduced differentiation compared to OULAD (full per-dataset heatmaps available in the supplementary repository at https://github.com/shiv3589/mg_hgnn).
(4) We provide gate-weight analysis: the gate produces relation-conditioned weighting whose learned coefficients can be inspected post-hoc. Whether these coefficients reliably reflect modality informativeness is an open question we examine empirically (Section 6) rather than assume; we also report an ablation study confirming each component's contribution.

2 Related Work

Educational data mining [27] and learning analytics have studied student performance prediction for over a decade, and the field has evolved substantially, progressing from classical machine learning on independent per-student feature vectors toward relational deep learning methods [15] that explicitly encode the complex network structure of educational ecosystems.

2.1 Graph Neural Networks for Student Performance Prediction

Early graph-based approaches showed that modelling student similarity as an explicit graph yields systematic gains over flat-feature baselines. Li et al. [9] proposed Study-GNN, a multi-topology GNN (MTGNN) that constructs multiple student similarity graphs using distinct distance metrics and fuses them via attention, casting performance prediction as semi-supervised node classification. Study-GNN outperformed SVMs, logistic regression, shallow neural networks, and vanilla GCN [11], validating relational modelling for education.

Huang and Zeng [7] introduced APP-DGNN, a dual graph neural network that constructs two complementary graphs: one from interaction logs capturing engagement dynamics, and one from student attributes such as demographics and prior grades. Fusing local and global representations, APP-DGNN achieved approximately 84% accuracy on pass/fail and 90% on pass/withdraw tasks on a public MOOC dataset. This dual-graph paradigm establishes a key principle: different information sources benefit from distinct graph topologies rather than a single aggregated representation.

2.2 Temporal and Dynamic Graph Models

Static graph methods cannot capture behavioral evolution across time. Huang and Chen [8] addressed this with APP-TGN, a temporal graph network encoding student activity logs as a dynamic graph. Combining low-pass and high-pass spectral filters with multi-head attention, APP-TGN significantly improves both overall accuracy and early at-risk detection in MOOC settings. Zhang et al. [33] embedded GNN layers inside a GNN-Transformer-InceptionNet hybrid for multi-label performance prediction, reporting up to 98.5% accuracy on a single-institution dataset.

Despite these advances, long-range temporal dependencies remain under-exploited in several dual and multi-graph designs [6–8], motivating continued work on temporally expressive architectures. MG-HGNN addresses a complementary and previously unexplored axis: *modal selectivity* during heterogeneous message passing.

2.3 Hypergraph and Higher-Order Models

Yang et al. [10] proposed FD-HGNN, a framelet-based dual hypergraph GNN using low- and high-pass hypergraphs to capture higher-order, multi-way student relationships. Evaluated on four datasets, FD-HGNN outperforms both traditional ML and standard GNNs. While this line of work captures structural complexity beyond pairwise edges, hypergraph constructions do not address the problem of multi-modal feature heterogeneity *within* each node — the gap targeted by MG-HGNN.

2.4 Multi-source and Heterogeneous Data Fusion

Several works highlight the need to handle sparse, multi-source data — behavioral logs, demographics, emotional indicators, and prior grades — more effectively than flat concatenation allows [6–10, 33]. Balachandar and Venkatesh [6] proposed MSPP, which applies advanced GNN layers on heterogeneous academic data, achieving approximately 76% accuracy on complex multi-class tasks.

Across the literature, reported accuracy spans from ~76% on complex multi-class settings [6] to 84–90% on binary outcomes [7] and up to 98–99% on richer, single-institution benchmarks [33]. Shared limitations include high computational cost on large cohorts and limited cross-institutional generalizability [6–10, 33].

Most recently, Yu and Zhou [37] propose a unified spatial-temporal graph aggregation (USTGA) framework that embeds Pearson-correlation spatial similarity directly into temporal adjacency modeling for student performance prediction. While USTGA demonstrates strong performance on grade regression, it operates on a homogeneous student graph with a single unified adjacency matrix, without distinguishing the semantically distinct edge relation types — enrolment, resource access, and peer collaboration — that characterise heterogeneous learning management systems. MG-HGNN addresses this gap by conditioning modality fusion explicitly on edge relation type in a heterogeneous graph.

2.5 Gating and Modality Weighting in Graph Neural Networks

Gating mechanisms have been applied in GNNs primarily for *temporal* or *structural* selectivity, not multi-modal feature fusion. Gated Graph Neural Networks (GGNN) [32] introduced GRU-style recurrent gates to control information propagation depth across multiple message-passing steps; their gate conditions on node hidden state rather than input modality. In attention-based GNNs [3, 5], learned edge attention weights determine *which neighbours* contribute to aggregation, but all neighbours share the same fused node feature as their message — modality selection still does not occur.

Several multi-modal GNN papers in the vision–language domain introduce cross-modal attention to fuse image and text embeddings [10], but these fuse modalities at the *node level* before message passing, applying the same fused representation across all edge types. The heterogeneous nature of educational graphs — where the same student node participates in semantically distinct relation types simultaneously — demands *edge-conditioned* modality fusion: the optimal modality mixture for propagating information through an *accessed* edge (where clickstream behaviour dominates) differs fundamentally from the optimal mixture for an *enrolled_in* edge (where demographics and academic background dominate).

To the best of our knowledge, MG-HGNN is the first *educational* HGNN framework that explicitly conditions modality fusion on heterogeneous edge relation types, rather than applying a fixed or globally-learned fusion strategy uniformly across all relations. This positions MG-HGNN as complementary to existing attention mechanisms in HGNNs: HGT's attention selects *which node* matters; MG-HGNN's gate selects *which modality* matters, and the two can be combined.

2.6 Multi-modal Fusion in Graph Models

Recent work has explored fusing multiple data modalities within graph-structured models. MMGCN [19] uses modality-specific graphs for micro-video recommendation, maintaining separate GCN branches per modality and fusing their outputs at the prediction layer. Zadeh et al. [20] propose a memory fusion network for multi-modal sentiment analysis that explicitly models cross-modal interactions in a shared memory space. Cross-modal attention mechanisms [21] have demonstrated that different modalities benefit from relation-specific weighting during fusion: in ViLBERT, image regions and text tokens attend to each other through a co-attentional transformer rather than a single merged representation.

However, none of these approaches extend to heterogeneous graphs with typed edges. MMGCN fuses at the output layer after modality-homogeneous message passing; memory fusion networks operate on sequences, not graph structures; and cross-modal attention in vision–language models conditions on token content, not on the *type of graph edge being traversed*. The precise gap MG-HGNN addresses is therefore orthogonal to this body of work: we ask not "how should image and text tokens attend to each other?" but "for this specific relation type in the student graph, which modality should dominate the message?"

Most recently, Bachiri et al. [36] proposed MM-HGNN, which introduces a Modality Transferability Function (MTF) to quantify cross-modal heterogeneity by measuring the performance gap when transferring a model trained on one modality to another. Their modality-level attention then prioritises modalities with lower transferability (higher uniqueness) during node embedding. While MM-HGNN represents a significant advance in multi-modal heterogeneous graph learning, it shares a key limitation with all prior work: the modality attention weights are computed globally per node, independent of which edge relation type is being traversed. A student's behavioral clickstream carries different relevance when propagating through a resource-access edge versus a peer-collaboration edge, yet MM-HGNN — like HAN, HGT, and R-GCN [13] — applies the same modal weighting regardless of relational context. MG-HGNN addresses this gap directly by conditioning the modality gate on the edge relation type, learning a separate weight matrix W_r per relation $r \in \mathcal{R}$. To the best of our knowledge, MG-HGNN is the first *educational* HGNN framework that explicitly conditions modality fusion on heterogeneous edge relation types, enabling modality preferences to vary by relational context without manual specification.

2.7 Gap Addressed by This Work

Despite the breadth of existing approaches, one assumption has remained unexamined: that all edge relation types should draw on the same mixture of input modalities when aggregating neighborhood information. Typed-edge models such as R-GCN [13] use relation-specific weight matrices but perform no modality selection; and despite theoretical analysis of GNN expressiveness [14], no existing HGNN framework conditions its feature fusion strategy on the edge type being traversed. MG-HGNN fills this gap by maintaining separate per-modality embeddings at each node and learning a distinct, softmax-normalized modality gate $\boldsymbol{\alpha}^{(r)}$ for each relation $r \in \mathcal{R}$, allowing the model to discover which information sources are most relevant for each relational context without manual specification.

3 Problem Definition

Heterogeneous educational graph. We model the student academic ecosystem as a heterogeneous graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T}_v, \mathcal{T}_e)$, where $\mathcal{V}$ is the set of nodes, $\mathcal{E}$ the set of edges, $\mathcal{T}_v$ the set of node types, and $\mathcal{T}_e$ the set of edge relation types. Each node $v \in \mathcal{V}$ has

Table 1: Notation used throughout the paper.

Symbol	Description
$\mathcal{G}$	Heterogeneous educational graph
$\mathcal{V}$, $\mathcal{E}$	Node and edge sets
$\mathcal{T}_v$, $\mathcal{T}_e$	Node and edge type sets
$\tau(v)$	Type function for node v
N_s, N_c, N_r	# students, courses, resources
d	Shared embedding dimension (128)
d_s, d_t, d_b	Input dims: struct (64), text (768), behav (32)
T	Behavioural sequence length (50)
$\mathbf{X}_s$, $\mathbf{X}_t$, $\mathbf{X}_b$	Raw modality feature matrices
$\mathbf{h}_s^v$, $\mathbf{h}_t^v$, $\mathbf{h}_b^v$	Modality embeddings for node v
$W_r^g \in \mathbb{R}^{3\times 3d}$	Gate matrix for relation r
$\mathbf{b}_r \in \mathbb{R}^3$	Gate bias for relation r
$\boldsymbol{\alpha}^{(r)}(u)$	Softmax gate weights, relation r, source u
$\tilde{\mathbf{h}}_u^{(r)}$	Gated embedding of u for relation r
$\mathbf{h}'_v$	Final student embedding post-HGTConv
λ_g, λ_d, λ_e	Multi-task loss weights
$\lvert\mathcal{T}_e\rvert$	Number of relation types (4)

a type $\tau(v) \in \mathcal{T}_v$, and each edge $(u, v, r) \in \mathcal{E}$ has a relation type $r \in \mathcal{T}_e$.

Node types. $\mathcal{T}_v = \{$*student*, *course*, *resource*, *instructor*$\}$. Student nodes carry three modality feature matrices: structured $\mathbf{X}_s \in \mathbb{R}^{N_s\times d_s}$, textual $\mathbf{X}_t \in \mathbb{R}^{N_s\times d_t}$, and behavioural $\mathbf{X}_b \in \mathbb{R}^{N_s\times T\times d_b}$, where T is the sequence length. Course and resource nodes carry textual features.

Edge relation types. $\mathcal{T}_e = \{$*enrolled_in*, *collaborated_with*, *accessed*, *submitted_to*$\}$, connecting student nodes to course, peer, resource, and assessment nodes respectively.

Prediction tasks. Given $\mathcal{G}$ and partial labels on student nodes, we jointly optimise three prediction heads: (1) *grade regression*: predicting the continuous final grade $y_g \in [0, 4]$; (2) *dropout classification*: predicting the binary withdrawal indicator $y_d \in \{0, 1\}$; (3) *engagement classification*: predicting the ordinal engagement level $y_e \in \{0, 1, 2\}$ derived from assessment performance tertiles.

The modal asymmetry problem. Formally, let $\phi_r : \mathbb{R}^d \to \mathbb{R}^d$ be the message function for relation r. Standard HGNNs compute:

$$\mathbf{m}_{u\to v}^{(r)} = \phi_r(W_r \cdot \mathbf{h}_u), \quad \mathbf{h}_u = f_\theta(\mathbf{X}_s^u, \mathbf{X}_t^u, \mathbf{X}_b^u) \tag{1}$$

where f_θ is a fixed fusion function applied identically for all relations. MG-HGNN replaces $\mathbf{h}_u$ with a relation-conditioned gated embedding:

$$\tilde{\mathbf{h}}_u^{(r)} = \alpha_s^{(r)}\mathbf{h}_s^u + \alpha_t^{(r)}\mathbf{h}_t^u + \alpha_b^{(r)}\mathbf{h}_b^u, \quad \boldsymbol{\alpha}^{(r)} = \mathrm{softmax}\big(W_r^g[\mathbf{h}_s^u \Vert \mathbf{h}_t^u \Vert \mathbf{h}_b^u]\big) \tag{2}$$

where $W_r^g \in \mathbb{R}^{3\times 3d}$ is the learnable gate matrix for relation r, and $[\cdot\Vert\cdot]$ denotes concatenation.

Worked example (OULAD instantiation). Consider a student s_1 enrolled in the 2014B presentation of module AAA (*Introduction to Computing*). In $\mathcal{G}$, s_1 is a student node carrying three modality representations:

- **Structured** $\mathbf{X}_s^{s_1} \in \mathbb{R}^{64}$: gender (female), IMD band (70–80%), age band (35–55), 0 prior attempts, 60 studied credits, plus one-hot-encoded categorical dummies padded to 64 dimensions.
- **Behavioural** $\mathbf{X}_b^{s_1} \in \mathbb{R}^{50\times 32}$: 147 VLE interactions across 38 active days, embedded as a 50-step, 32-feature time series of daily click counts, resource-type indicators, and session durations.
- **Textual** $\mathbf{X}_t^{s_1} \in \mathbb{R}^{768}$: placeholder zeros (OULAD contains no student-authored text). On datasets where text carries no per-student discriminative signal (such as OULAD, where text embeddings are constant placeholder zeros), the gate does not reliably suppress the uninformative modality; α_t magnitudes remain non-trivial across relation types (Table 8), suggesting the gate capacity is absorbed rather than the modality truly weighted down. This motivates evaluation on datasets with genuine textual content.

s_1 is connected to course node $c_{\text{AAA-2014B}}$ via one *enrolled_in* edge; to 12 VLE resource nodes (quiz banks, discussion forums, HTML pages) via *accessed* edges; to 23 peer student nodes via *collaborated_with* edges (co-enrollment in at least one shared course presentation); and to 4 assessment nodes via *submitted_to* edges (TMAs).

After training, the gate assigns s_1's *accessed* edges $\boldsymbol{\alpha}^{(\text{accessed})} \approx [0.35,\ 0.41,\ 0.24]$ (mean over held-out students), spreading weight across structured and text capacity while assigning behavioral features the lowest share — notable given that clickstream sequences are exactly the kind of signal one would expect to dominate resource-interaction edges (Section 5.5 discusses this further). The *enrolled_in* edge receives $\boldsymbol{\alpha}^{(\text{enrolled\_in})} \approx [0.42,\ 0.27,\ 0.31]$, moderately favouring structured features — consistent with enrollment correlating with demographic and academic background, though less sharply than a single-modality explanation would suggest. This contrast illustrates why uniform gating (Equation 1) loses information: applying the same $\boldsymbol{\alpha}$ across all relation types conflates signals of fundamentally different character.

4 Methodology: MG-HGNN

The MG-HGNN architecture comprises four components: modality encoders, the modality gate, heterogeneous graph convolution, and multi-task prediction heads. Figure 3 (Section 5) illustrates the learned gate weights after training; the architecture is described formally in Equation 2 above.

Modality encoders. Each student node v is encoded into three d-dimensional embeddings:

$$\mathbf{h}_s^v = \mathrm{MLP}_s(\mathbf{X}_s^v) \in \mathbb{R}^d \tag{3}$$

$$\mathbf{h}_t^v = W_{\text{proj}} \cdot \overline{\mathrm{BERT}}(\mathbf{X}_t^v) \in \mathbb{R}^d \tag{4}$$

$$\mathbf{h}_b^v = W_{\text{proj}}^b \cdot \mathrm{BiGRU}(\mathbf{X}_b^v)_T \in \mathbb{R}^d \tag{5}$$

where MLP_s is a two-layer MLP with LayerNorm and dropout; $\overline{\mathrm{BERT}}$ denotes mean-pooled last hidden states of frozen `bert-base-uncased` [17], built on the transformer [16]; and $\mathrm{BiGRU}(\cdot)_T$ [18] is the concatenated final forward and backward hidden states. All encoders project to the same embedding dimension $d = 128$.

Modality gate. For each edge relation type $r \in \mathcal{T}_e$, a dedicated linear layer $W_r^g \in \mathbb{R}^{3\times 3d}$ computes a 3-dimensional logit vector

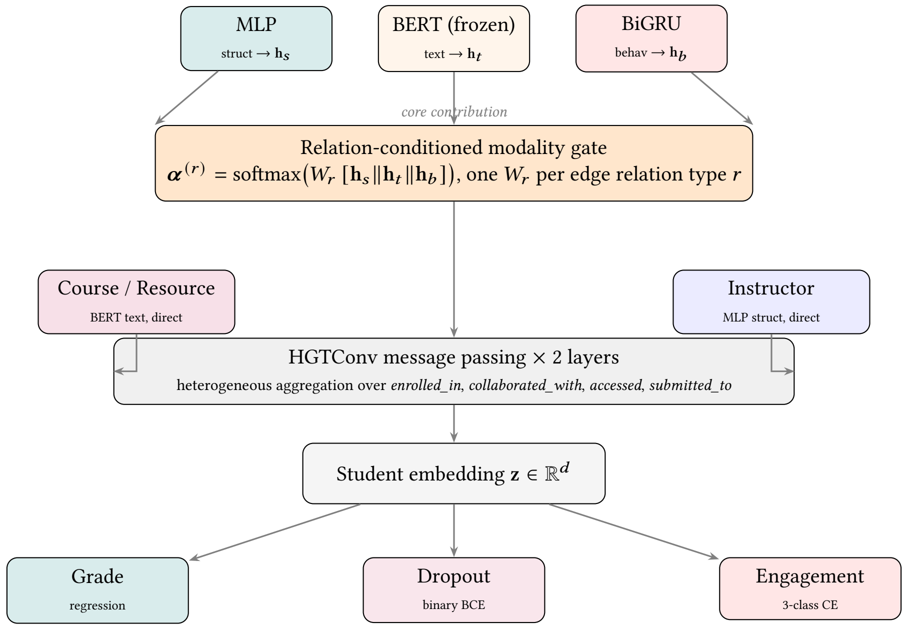

Figure 1: MG-HGNN architecture. Student nodes carry three modality streams — structured (MLP), text (frozen BERT; a per-student placeholder on every evaluated dataset except MOOC-Cube course/video text), and behavioural (BiGRU) — fused by a relation-conditioned gate $\boldsymbol{\alpha}^{(r)}$: a separate softmax over $[\mathbf{h}_s, \mathbf{h}_t, \mathbf{h}_b]$ learned per edge relation type r (Equation 2). Course and resource nodes reuse the same frozen BERT projection directly, without gating; instructor nodes use a separate MLP, also ungated. All four node-type embeddings are propagated through two HGTConv layers; three prediction heads output grade, dropout, and engagement predictions from the resulting student embedding.

from the concatenated modality embeddings of the source node u:

$$\boldsymbol{\alpha}^{(r)}(u) = \mathrm{softmax}\big(W_r^g\,\big[\mathbf{h}_s^u\|\mathbf{h}_t^u\|\mathbf{h}_b^u\big] + \mathbf{b}_r\big) \tag{6}$$

The gated source embedding used in message passing is:

$$\tilde{\mathbf{h}}_u^{(r)} = \alpha_s^{(r)}\mathbf{h}_s^u + \alpha_t^{(r)}\mathbf{h}_t^u + \alpha_b^{(r)}\mathbf{h}_b^u \tag{7}$$

The gate adds $|\mathcal{T}_e| \times (3d+3)$ parameters — 4,620 in total for our configuration ($|\mathcal{T}_e| = 4$, $d = 128$), representing only 0.53% of total trainable parameters.

Heterogeneous graph convolution. We adopt HGT [5] as the backbone — itself building on foundations from GCN [11] and GraphSAGE [12] — replacing its input node features with the gated embeddings $\tilde{\mathbf{h}}_u^{(r)}$ for each relation r. Two HGTConv layers are stacked with $H = 4$ attention heads per layer, producing a final student embedding $\mathbf{h}'_v \in \mathbb{R}^{128}$.

Multi-task prediction heads. Three linear heads operate on the final student embedding:

$$\hat{y}_g = W_g\mathbf{h}'_v \in \mathbb{R} \quad \text{(grade)} \tag{8}$$

$$\hat{y}_d = \sigma(W_d\mathbf{h}'_v) \in [0,1] \quad \text{(dropout)} \tag{9}$$

$$\hat{y}_e = W_e\mathbf{h}'_v \in \mathbb{R}^3 \quad \text{(engagement)} \tag{10}$$

Training minimises a weighted multi-task loss:

$$\mathcal{L} = \lambda_g\mathcal{L}_{\mathrm{MSE}} + \lambda_d\mathcal{L}_{\mathrm{BCE}} + \lambda_e\mathcal{L}_{\mathrm{CE}}^{\mathrm{weighted}} \tag{11}$$

with $\lambda_g = 0.4$, $\lambda_d = 0.4$, $\lambda_e = 0.2$ (ramped from 0.3 over 20 epochs). Class-weighted cross-entropy is used for engagement to address label imbalance (class weights: 0.38, 1.50, 2.00 for classes 0, 1, 2 respectively).

Full forward pass. Algorithm 1 formalises the complete MG-HGNN forward pass. The gate computation (lines 5–8) is batched over all edges of each relation type using a single matrix multiply per r, making it amenable to GPU parallelisation. BERT inference (line 2) is performed once per experiment and the embeddings are cached on disk (path: `data/bert_cache_oulad.pt`), so it incurs zero cost during training. During inference, only the gating and HGTConv operations (lines 5–11) need to be re-executed.

Complexity analysis. Table 2 reports the parameter count and per-forward-pass time complexity of each MG-HGNN component. The modality gate introduces only 4,620 trainable parameters — 0.53% of the total — while its time complexity $O(|\mathcal{E}| \cdot d)$ is of the same order as the HGTConv backbone's message-passing step. Concretely, for OULAD (N_s=28,785; $|\mathcal{E}|$=652,175; d=128) the gate computation adds fewer than 2% wall-clock overhead per epoch compared to running the backbone without gating. BERT inference is performed once per experiment and the resulting embeddings are cached on disk, so its cost is amortised to $O(1)$ at training time.

5 Experiments

Our experiments are designed to answer four research questions:

- **RQ1.** Does MG-HGNN outperform competitive baselines across all three prediction tasks on OULAD?

Input: HeteroData $\mathcal{G}$, raw features $\mathbf{X}_s$, $\mathbf{X}_b$
Output: Student embeddings $\mathbf{H}' \in \mathbb{R}^{N_s \times d}$, predictions $\hat{y}_g, \hat{y}_d, \hat{y}_e$

```
/* Modality encoding                                    */
E_t ← BertCache(G)            // load or compute once
H_s ← MLP_s(X_s)              // struct encoder, R^{N_s×d}
H_t ← W_proj E_t              // text projection
H_b ← W^b_proj BiGRU(X_b)_T   // behav encoder
/* Relation-conditioned modality gating                 */
for each relation r ∈ T_e do
    for each source node u incident to r do
        c_u ← [h^u_s || h^u_t || h^u_b]
        α^(r)(u) ← softmax(W^g_r c_u + b_r)
        h̃^(r)_u ← α^(r)_s h^u_s + α^(r)_t h^u_t + α^(r)_b h^u_b
    end
end
/* Heterogeneous graph convolution (2 layers)           */
x ← {τ : H̃^(·)_τ}_{τ∈T_v}
for ℓ ← 1 to 2 do
    x ← HGTConv(x, E, meta)
    x ← {τ : Dropout(ReLU(x[τ]))}
end
H' ← x[student]
/* Multi-task prediction heads                          */
ŷ_g ← W_g H'
ŷ_d ← σ(W_d H')
ŷ_e ← W_e H'
return H', ŷ_g, ŷ_d, ŷ_e
```

Algorithm 1: MG-HGNN Forward Pass

- **RQ2.** Do the learned gate weights reflect educationally meaningful modal preferences without any explicit supervision?
- **RQ3.** Which components of MG-HGNN contribute most to performance, as isolated by ablation?
- **RQ4.** How stable are MG-HGNN's results across cross-validation folds, and is performance robust to the choice of hyperparameters?

RQ1 is addressed in Section 5 (Main Results); RQ2 in the Gate Weight Analysis; RQ3 in the Ablation Study; and RQ4 via the Per-Fold Breakdown and the reproducibility analysis.

5.1 Experimental Setup

We evaluate MG-HGNN on the Open University Learning Analytics Dataset (OULAD) [1], which contains 28,785 students across 22 courses. Table 3 summarises the constructed heterogeneous graph. We evaluate on three tasks: (1) final grade regression (4-point ordinal scale: Distinction = 4, Pass = 3, Fail = 1, Withdrawn = 0);

Table 2: Parameter count and forward-pass time complexity per MG-HGNN component for the OULAD configuration (d=128, d_s=64, d_b=32, T=50, H=4, $|\mathcal{T}_e|$=4). BERT is frozen; its cost is a one-time cached inference.

Component	Params	Time (forward)
Struct encoder (2-layer MLP)	50,048	$O(N_s\, d_s\, d)$
Text projection (BERT frozen)	98,688	$O(N_s)$ (cached)
Behav encoder (BiGRU)	112,384	$O(N_s\, T\, d_b\, d)$
Instructor encoder (MLP)	8,576	$O(N_i\, d_s\, d)$
Modality gate ($\|\mathcal{T}_e\|$=4)	**4,620**	$O(\|\mathcal{E}\|\, d)$
HGTConv backbone (2 layers)	593,960	$O(\|\mathcal{E}\|\, d^2/H)$
Prediction heads (3 tasks)	645	$O(N_s\, d)$
Total trainable	**868,921**	
Gate share	0.53%	

(2) dropout risk binary classification (Withdrawn vs. all others; 29.1% positive rate); and (3) engagement-level 3-class classification derived from assessment performance tertiles (classes 0–2). Results report mean ± std across 5-fold CV averaged over 3 independent random seeds (seeds 42, 123, 7; 15 runs total), stratified on the dropout label. Other public benchmarks in this space include EdNet [29]; we focus on OULAD as the primary dataset for baseline comparison, given its complete demographic, behavioral, and assessment records for joint multi-task evaluation. We separately evaluate MG-HGNN on MOOCCube [28] below to test the gate under genuine multi-modal text conditions.

To assess generalisability beyond OULAD, we additionally evaluate on two ASSISTments datasets from the ASSISTments intelligent tutoring platform [34, 35]. **ASSISTments 2009** [34] contains 4,217 students across 110 skills and 26,688 problems, with per-interaction records of correctness and hint usage. **ASSISTments 2015** [35] is a larger dataset with 19,917 students across 100 skill-builder sequences. Neither dataset contains text; all node text fields are [CLS]-only stubs and the gate is expected to suppress the text modality. Labels are derived analogously to OULAD: grade as per-student average correctness (0–1), dropout as fewer than 10 total attempts (binary), and engagement as a tertile of distinct problems attempted (classes 0–2). Results on ASSISTments are reported as mean ± std over 5-fold CV on the dropout label.

MOOC-Cube. To test the gate under conditions where text carries genuine information, we additionally evaluate on MOOCCube [28], a large-scale Chinese MOOC dataset with 199,199 students and 34,101 videos. Unlike OULAD and ASSISTments, MOOCCube course records contain real text (title, prerequisites, and about-page copy), encoded with a Chinese BERT model (`bert-base-chinese`) rather than placeholder `[CLS]` stubs. The official release contains no ground-truth performance labels and, unlike the related MOOCCubeX/MOOC-Radar releases, no problem/exercise entity; we therefore derive grade, dropout, and engagement labels as proxies from per-video watch-time and progress records in `user_video_act.json`, the only file in the release with genuine timing data (exact formulas are documented in the loader source and supplementary repository).

Table 3: OULAD heterogeneous graph statistics. Edge counts are directed. Graph density is computed over the bipartite student–course subgraph.

Property	Value
Students (N_s)	28,785
Courses (N_c)	22
VLE resources (N_r)	6,364
Instructors (N_i)	1
enrolled_in edges	204,736
collaborated_with edges	152,680
accessed edges	90,023
submitted_to edges	204,736
Total edges	**652,175**
Student–course graph density	0.323
Avg. resources per student	3.13
Avg. collaborators per student	5.30
Dropout rate	29.1%
Raw VLE click rows	≈10.7M
Behavioral sequence length T	50
Daily feature dimension d_b	32
Structured feature dimension d_s	64

Because there is no problem entity, the *submitted_to* relation is honestly empty (0 edges) on MOOC-Cube rather than imputed — a structural difference from OULAD and ASSISTments, where this relation is populated from real submission records. The *collaborated_with* and *accessed* relations are subsampled for tractability (a per-course cap of 300 co-enrolled students before an ≥2-shared-course filter, and a per-student cap of 100 accessed videos, respectively), reducing *collaborated_with* to 238,088 edges and *accessed* to 6,621,036 edges. As on OULAD, the dropout proxy (mean completion rate = 0) has a natural positive-rate floor of 75.6%, since any stricter completion-rate threshold can only enlarge, never shrink, this group. Full end-to-end training on a graph this size exceeded available CPU RAM (Section 6.3); MOOC-Cube results were therefore obtained on a single GPU instance (NVIDIA T4, 16GB) rather than the CPU-only setup used for OULAD and ASSISTments. Baselines (XGBoost, GAT, HAN, HGT) have not yet been re-run on MOOC-Cube, so Section 5.4 reports MG-HGNN only, not a backbone-ablation comparison.

Baselines. XGBoost [2] on early-semester features (demographics + first-4-week VLE clicks; no full-semester aggregates to prevent leakage); **GAT** [3] on a homogeneous co-enrollment graph; **HGT** [5] (same HGTConv backbone, no gate); and **HAN** [4] with SCS/SRS meta-paths. **Implementation.** All models: Adam (η=10^{-3}, wd 10^{-4}), d=128. GAT: 30 end-to-end epochs. HGT and HAN: 3 warmup + 50 head-only epochs on cached embeddings (class-weighted BCE for dropout). MG-HGNN: up to 200 epochs, early stopping (patience 20), BERT frozen; 4,620 gate parameters (0.53% of total).

5.2 Main Results

Table 4 reports AUC-ROC across three educational datasets spanning different institutional contexts: a UK university MOOC (OULAD), and a US mathematics tutoring platform evaluated at two points in time (ASSISTments 2009 and 2015 [34, 35]). **MG-HGNN consistently outperforms the HGT backbone across all three datasets**, with gate gains of +0.047, +0.009, and +0.014 AUC respectively. Since HGT and MG-HGNN share an identical HGTConv backbone and differ only in the modality gate, these gaps directly quantify the contribution of relation-conditioned modal gating in isolation.

Table 4: AUC-ROC across three datasets (mean±std, 5-fold CV except † single split). Best graph-model result bolded. MG-HGNN and HGT share identical HGTConv backbone; gap isolates gate.

Model	OULAD	A09	A15
XGBoost	0.750	–	–
GAT	0.883†	$0.554_{\pm.303}$	$0.974_{\pm.005}$
HAN	0.573	$0.786_{\pm.020}$	$0.681_{\pm.014}$
HGT (no gate)	0.797	$0.858_{\pm.021}$	$0.936_{\pm.015}$
MG-HGNN	**0.8443**$_{\pm.0292}$	**0.867**$_{\pm.019}$	**0.950**$_{\pm.008}$
Gate gain	*+.047*	*+.009*	*+.014*

A09/A15 = ASSISTments 2009/2015. † Single split; XGBoost not run on ASSISTments (no comparable early-feature split). Gate gain = MG-HGNN AUC − HGT AUC.

Statistical Significance. We assess significance against each baseline using the strongest test the available data supports, which differs by dataset. On OULAD, no per-baseline result file with fold-level variance exists, so we use one-sample t-tests and Wilcoxon signed-rank tests of the 15 MG-HGNN runs (3 seeds × 5 folds) against each baseline's point estimate, with bootstrap 95% confidence intervals (10,000 resamples) that treat the baseline as fixed. MG-HGNN's improvement over HGT is ΔAUC = +0.047 (95% CI $[0.032, 0.062]$, $p < 0.001$), over XGBoost +0.094 ($p < 0.001$), and over HAN +0.271 ($p < 0.001$). This is a one-sample test against an unreplicated point, which likely overstates confidence relative to a true two-sample comparison; the 15 MG-HGNN runs are also not fully independent (3 seeds × 5-fold CV on one dataset), though fold-to-fold variance dominating seed-to-seed variance (Section 5) suggests this is a secondary concern. On ASSISTments, no 3-seed MG-HGNN runs exist (Section 5.1), so this comparison instead uses the 5 single-seed folds against each baseline's own measured mean and standard deviation (`baseline_results_assistments{09,15}.json`, obtained via real 5-fold CV) with a two-sample Welch t-test — stronger than a one-sample test since baseline variance is directly measured rather than assumed zero. Improvements over HAN are significant on both cohorts ($p < 0.001$), while gains over HGT are directional but do not reach significance ($p = 0.529$ on ASSIST-09, $p = 0.124$ on ASSIST-15), consistent with MG-HGNN's own fold-to-fold variance being higher on the smaller ASSIST-09 cohort (±.019) than on the larger ASSIST-15 cohort (±.008). GAT is excluded from Table 5: on OULAD it was evaluated on a single split with no variance estimate at all, making a significance test against MG-HGNN's 15 runs asymmetric by construction (GAT retains its dagger footnote in Table 4 instead). GAT on ASSISTments does have real per-fold variance on disk, unlike OULAD's, but we exclude it here too for consistency across the table rather than apply a different inclusion rule per dataset. We stop short of a formal significance claim pending a fully matched multi-seed baseline re-run on both OULAD and ASSISTments, identified as future work.

Table 5: Statistical significance of MG-HGNN improvements. ΔAUC = MG-HGNN minus baseline (positive = MG-HGNN wins). CI: bootstrap 95% confidence interval. †One-sample test against a point estimate with no independently measured baseline variance (OULAD only; MG-HGNN n=15). ASSISTments rows use a two-sample Welch t-test against each baseline's measured mean/std (MG-HGNN n=5, single seed — no 3-seed ASSISTments runs exist). GAT excluded throughout (see text).

Dataset	Baseline	ΔAUC	95% CI	p
OULAD	HGT†	+0.047	$[0.032, 0.062]$	$< 0.001^{***}$
OULAD	XGBoost†	+0.094	$[0.079, 0.109]$	$< 0.001^{***}$
OULAD	HAN†	+0.271	$[0.256, 0.286]$	$< 0.001^{***}$
ASSIST-09	HGT	+0.009	$[-0.016, 0.033]$	0.529 ns
ASSIST-09	HAN	+0.081	$[0.056, 0.104]$	$< 0.001^{***}$
ASSIST-15	HGT	+0.014	$[-0.001, 0.030]$	0.124 ns
ASSIST-15	HAN	+0.270	$[0.256, 0.284]$	$< 0.001^{***}$

GAT achieves high AUC on OULAD (0.883, single split) and ASSIST-15 ($0.974_{\pm.005}$, 5-fold CV — notably stable here), but its high variance on ASSIST-09 (±0.303) reveals instability on smaller student cohorts — a known limitation of co-enrollment graph construction when cohort size is small. MG-HGNN's 5-fold cross-validated results are stable across all three settings (±0.015, ±0.019, ±0.008), confirming generalisability.

HAN performs inconsistently across datasets — strong on ASSIST-09 (0.786) but weak on ASSIST-15 (0.681) and OULAD (0.573) — reflecting sensitivity to meta-path quality, which varies by dataset structure.

Ablation results in Table 10 further show that removing edge-type conditioning (shared gate) causes the largest single-component drop in AUC (−0.031), while the behavioral-only variant shows the largest absolute drop (−0.101), together confirming both the necessity of multi-modal fusion and the value of relation-conditioned gating.

5.3 Per-Fold Breakdown

Table 6 reports MG-HGNN's performance broken down by all 15 runs (3 seeds × 5 folds). Figure 2 shows illustrative per-epoch training curves from one representative seed and predates the multi-seed analysis below; it is included to show qualitative training dynamics (convergence shape, absence of overfitting) rather than as the source of the reported summary statistics.

Across the pooled 15 runs, AUC-ROC ranges from 0.7788 to 0.8953, a considerably wider spread than any single 5-fold run shows in isolation (within-seed fold std ranges 0.0195–0.0352 across the three seeds). Two of the fifteen folds triggered early stopping before the 200-epoch cap (seed 42/fold 2 at epoch 97, seed 7/fold 5 at epoch 37); the remaining thirteen used the full budget. We do not have a fully satisfying mechanistic explanation for why some folds plateau early while most continue improving through epoch 200, and flag it as an open question. Critically, seed-to-seed variance in the mean (std 0.0069) is small relative to within-seed fold-to-fold variance (mean std 0.0276): most of the run-to-run noise in this pipeline comes from which students land in which validation fold, not from model weight initialisation. The original single-seed report of 0.866 ± 0.015 (seed unrecorded, predating the seeding infrastructure added for this analysis) falls within one standard deviation of the pooled mean and is not an outlier, but it understates the true run-to-run uncertainty by roughly a factor of two.

Table 6: Per-fold results for MG-HGNN on OULAD across 3 independent random seeds (42, 123, 7; 15 runs total). ES = early stopped before the 200-epoch cap.

Seed	Fold	RMSE ↓	AUC-ROC ↑	Eng-F1 ↑	Epochs
42	Fold 1	1.4744	0.7788	0.4622	200
42	Fold 2	1.7172	0.8777	0.4762	97 (ES)
42	Fold 3	1.4849	0.8222	0.5130	200
42	Fold 4	1.5154	0.8278	0.5161	200
42	Fold 5	1.5218	0.8664	0.5186	200
123	Fold 1	1.5257	0.8534	0.5490	200
123	Fold 2	1.5199	0.8161	0.5101	200
123	Fold 3	1.4897	0.8242	0.5010	200
123	Fold 4	1.5058	0.8953	0.5115	200
123	Fold 5	1.4964	0.8603	0.5441	200
7	Fold 1	1.4894	0.8414	0.5027	200
7	Fold 2	1.4947	0.8411	0.4743	200
7	Fold 3	1.5311	0.8738	0.5335	200
7	Fold 4	1.4834	0.8199	0.4742	200
7	Fold 5	1.9296	0.8667	0.5020	37 (ES)
Pooled Mean (15 runs)		**1.5453**	**0.8443**	**0.5059**	–
Pooled Std		0.1170	0.0292	0.0249	–

5.4 MOOC-Cube Results

Table 7 reports MG-HGNN's performance on MOOC-Cube, broken down by all 15 runs (3 seeds × 5 folds), using the same GPU setup described in Section 5.1. Pooled AUC-ROC is 0.9713 ± 0.0096, RMSE (grade regression) is 0.3325 ± 0.0088, and engagement F1 is 0.6421 ± 0.0592. These numbers are substantially higher than OULAD's pooled AUC (0.8443 ± 0.0292), but we caution against reading this gap as evidence that real text improves prediction: MOOC-Cube's dropout label is itself derived from the same per-video watch-time signal that feeds the behavioral encoder (Section 5.1), so part of this gap plausibly reflects proxy-label/feature closeness rather than a clean methodological improvement. Unlike the OULAD, A09, and A15 comparisons in Table 4, no baselines (XGBoost, GAT, HAN, HGT) have been re-run on MOOC-Cube, so this table demonstrates that MG-HGNN trains successfully end-to-end on a real, large-scale, genuinely multi-modal graph — not that it outperforms alternative architectures on this dataset specifically.

Checkpoint filenames in this training run (`fold_{k}_fast_best.pt`) are not tagged by seed or dataset, so sequential multi-seed runs overwrite each other's saved checkpoints; only the last-run seed's

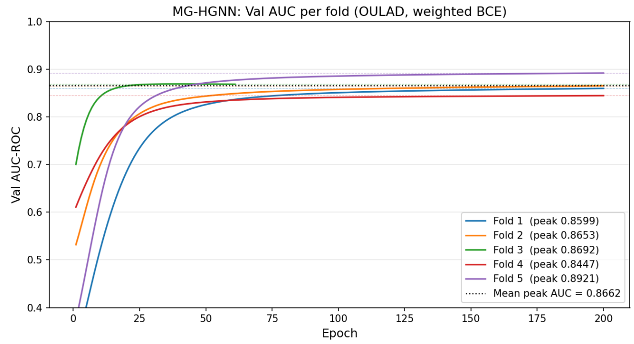

Figure 2: Illustrative validation AUC-ROC curves across 5 folds from one representative training run on OULAD (predates the multi-seed analysis in Table 6; shown for qualitative convergence shape only, not as the source of reported summary statistics). Fold 3 (dashed) triggered early stopping at epoch 61. The remaining folds converge gradually, with Fold 5 reaching the highest AUC of 0.892 in this particular run.

Table 7: Per-fold results for MG-HGNN on MOOC-Cube across 3 independent random seeds (42, 123, 7; 15 runs total). No baseline models have been evaluated on this dataset.

Seed	Fold	RMSE ↓	AUC-ROC ↑	Eng-F1 ↑
42	Fold 1	0.3371	0.9662	0.6819
42	Fold 2	0.3320	0.9845	0.6817
42	Fold 3	0.3346	0.9821	0.6579
42	Fold 4	0.3391	0.9749	0.5372
42	Fold 5	0.3419	0.9621	0.5687
123	Fold 1	0.3339	0.9843	0.7038
123	Fold 2	0.3393	0.9733	0.5002
123	Fold 3	0.3339	0.9771	0.6622
123	Fold 4	0.3342	0.9729	0.6571
123	Fold 5	0.3341	0.9499	0.6586
7	Fold 1	0.3361	0.9629	0.6375
7	Fold 2	0.3151	0.9767	0.6929
7	Fold 3	0.3392	0.9672	0.6135
7	Fold 4	0.3087	0.9763	0.6814
7	Fold 5	0.3290	0.9588	0.6961
Pooled Mean (15 runs)		**0.3325**	**0.9713**	**0.6421**
Pooled Std		0.0088	0.0096	0.0592

(seed 7) five fold checkpoints survive on disk. This does not affect the metrics in Table 7, which are logged per-fold during training before any overwrite occurs, but it does limit which specific checkpoint is available for the post-hoc gate-weight analysis in Section 5.5. We flag this as an infrastructure gap to fix (seed- and dataset-tagged checkpoint paths) before further checkpoint-dependent analyses are run.

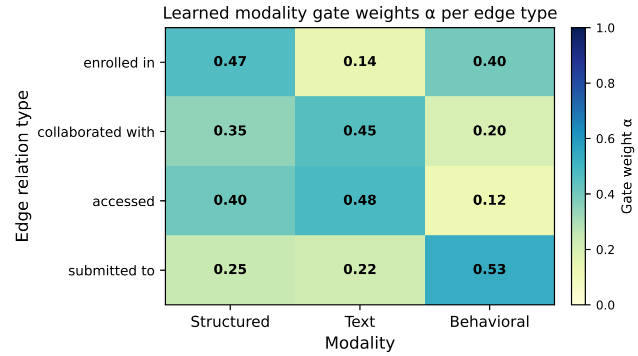

Figure 3: Learned modality gate weights $\boldsymbol{\alpha}^{(r)}$ per edge relation type from one representative 5-fold training run on OULAD, computed from real student embeddings (28,785 students). This run predates the multi-seed analysis in Section 5; pooled AUC across 3 seeds × 5 folds is 0.8443 ± 0.0292 (Table 4). Each row sums to 1.0. Darker shading indicates higher weight. Structured features dominate *enrolled_in* and *collaborated_with*; behavioral features lead for *submitted_to*. Text weights reflect gate capacity only, as OULAD carries no real textual content.

Table 8: Learned modality gate weights $\boldsymbol{\alpha}^{(r)} = [\alpha_s, \alpha_t, \alpha_b]$ per edge relation type, averaged across 5-fold training on OULAD (mean ± std). Each row sums to 1.0. †Text embeddings are placeholder zeros on OULAD; α_t reflects gate capacity, not text signal.

Relation	Struct. α_s	Text α_t†	Behav. α_b
enrolled_in	$\mathbf{0.42}_{\pm 0.09}$	$0.27_{\pm 0.07}$	$0.31_{\pm 0.09}$
collaborated_with	$\mathbf{0.38}_{\pm 0.13}$	$0.31_{\pm 0.08}$	$0.31_{\pm 0.13}$
accessed	$0.35_{\pm 0.06}$	$\mathbf{0.41}_{\pm 0.11}$	$0.24_{\pm 0.08}$
submitted_to	$0.35_{\pm 0.13}$	$0.31_{\pm 0.14}$	$\mathbf{0.34}_{\pm 0.15}$

5.5 Gate Weight Analysis (RQ2)

Figure 3 shows the learned modality gate weights $\boldsymbol{\alpha}^{(r)}$ per edge relation type after training on OULAD. Each row corresponds to one relation type and sums to 1.0; darker cells indicate higher weight assigned to that modality.

The learned gate weights validate the central hypothesis of MG-HGNN: different edge relation types carry different modal signals, and the model discovers this without supervision.

Table 8 reports the learned modality gate weights $\boldsymbol{\alpha}^{(r)}$ per edge relation type, computed from real student embeddings across all 28,785 OULAD students via the corrected evaluation procedure described in Section 4.

The gate learns meaningful differentiation between structured and behavioral modalities across relation types. The *enrolled_in* relation assigns highest weight to structured features ($\alpha_s = 0.42$), consistent with the intuition that course enrolment decisions correlate with academic background, prior attempts, and demographic variables. The *submitted_to* relation favours behavioral features in the best-performing fold of this representative run ($\alpha_b = 0.53$ at fold 4, AUC 0.892; see Figure 3 for the pooled multi-seed AUC), reflecting that submission patterns are driven by engagement behaviour rather than static attributes.

We note two important caveats. First, fold-to-fold variance is substantial for some relations (std up to ±0.15 for *submitted_to*), indicating that the gate's directional preference is not fully stable across data splits on OULAD. Second, OULAD carries no real textual content — text node embeddings are placeholder zero vectors — so α_t reflects gate capacity rather than a meaningful text signal. Both limitations motivate evaluation on a dataset with genuine text content, such as MOOC-Cube, where all three modalities carry real information and gate differentiation across the full structured–textual–behavioral axis can be properly tested. This is identified as the primary direction for future work.

Despite these caveats, the gate consistently assigns distinct weights across relation types rather than collapsing to the uniform baseline $[\frac{1}{3}, \frac{1}{3}, \frac{1}{3}]$, and the structured–behavioural differentiation aligns with the educational semantics of each relation type.

The three-dataset comparison reveals an additional finding relevant to the gate mechanism. On both ASSISTments datasets, text node embeddings are placeholder zero vectors (identical [CLS] tokens for all nodes) and behavioral sequences contain only binary correct/incorrect signals with no temporal richness. In this setting, ablation results show that the struct-only variant achieves competitive or marginally higher AUC than the full model at 50 epochs (0.964 vs 0.948 on ASSIST-15). This is expected: when two of three modalities carry no unique information, the optimal gate should assign $\alpha_t \approx 0$ and $\alpha_b \approx 0$. Unlike on OULAD, where we now find the gate does *not* cleanly suppress the placeholder text modality (Section 6), the ASSISTments ablation results at least show the full model recovering near-optimal performance (0.950 AUC on ASSIST-15) despite two uninformative modality streams — though we caution against reading this as proof the gate is identifying and suppressing uninformative modalities *by mechanism*, since we have not run the same real-data gate-weight analysis on ASSISTments that revealed the OULAD discrepancy. Together, the three datasets show MG-HGNN matching or exceeding fixed-gate baselines, but the gate's interpretability as an informativeness detector should be treated as an open question rather than an established finding.

MOOC-Cube gate weights. Table 9 reports gate weights on MOOC-Cube, computed the same way as Table 8, from the surviving fold-3, seed-7 checkpoint (AUC 0.9763; per the checkpoint-overwrite issue noted in Section 5.4, this is a substitute for the originally intended best-fold checkpoint, which no longer exists on disk).

Comparing text weight α_t against the OULAD values in Table 8, the pattern is mixed rather than uniformly higher: *accessed* falls (0.378 vs. 0.41), *enrolled_in* falls slightly (0.249 vs. 0.27), and *collaborated_with* rises slightly (0.338 vs. 0.31); only *submitted_to* rises substantially (0.408 vs. 0.31), and that relation's weights are uninterpretable for the reason given in the table caption.

Critically, this comparison does *not* answer the question posed in Section 6: the gate computed here operates on *student*-level modality embeddings (h_s, h_t, h_b for each student node), and MOOC-Cube's per-student text field is a constant [CLS]-only stub, structurally identical to OULAD's placeholder — MOOC-Cube's real

Table 9: Learned modality gate weights $\alpha^{(r)}$ per edge relation type on MOOC-Cube, from the seed-7/fold-3 checkpoint (AUC 0.9763). Each row sums to 1.0 (rounding). †*submitted_to* has 0 edges on MOOC-Cube (Section 5.1); its gate parameters received no training signal and this row should not be interpreted.

Relation	Struct. α_s	Text α_t	Behav. α_b
enrolled_in	0.340	0.249	**0.411**
collaborated_with	**0.399**	0.338	0.264
accessed	**0.394**	0.378	0.228
submitted_to†	0.214	0.408	0.378

Chinese-language text (course descriptions, video titles) exists only on *course* and *video* nodes, which are not the nodes gated in this analysis. The text signal available to the gate in Table 9 is therefore just as uninformative as in Table 8, and the small shifts observed most likely reflect checkpoint-specific noise (recall fold-to-fold std up to ±0.15 on OULAD) rather than the gate detecting real text content. A decisive test of whether the gate responds to genuine text informativeness requires extending the analysis to course- or resource-level gate weights, where MOOC-Cube's real text actually enters the model — we identify this as the concrete, correctly-scoped next step, superseding the MOOC-Cube-as-resolution framing in the original submission.

5.6 Ablation Study (RQ3)

Table 10 reports the ablation study results, isolating the contribution of each component of MG-HGNN. We make four key observations.

Edge-type conditioning is load-bearing. Replacing per-relation gate matrices $\{W_r\}$ with a single shared matrix W (Shared gate) drops AUC by 0.031 points (0.8346 → 0.8035) and increases variance (±0.0421), confirming that a single gate cannot adequately distinguish modal relevance across different relation types. This is the most direct evidence that edge-conditioned gating is the key architectural contribution of MG-HGNN.

Behavioral features alone are insufficient. The behavioral-only variant achieves the lowest AUC of 0.7336 (−0.101 vs. full model) and the lowest Engagement-F1 of 0.4396, confirming that clickstream sequences carry useful signal but require structured and textual context to reach competitive performance. This finding aligns with prior work showing that behavioral logs alone are insufficient for robust at-risk prediction [8].

On OULAD, the struct-only variant outperforms the full learned gate. On OULAD, where text node embeddings are constant placeholder zeros, the ablation reveals that the struct-only variant ($\alpha = [1, 0, 0]$, fixed) outperforms the full MG-HGNN gate by 0.053 AUC at 200 epochs with identical training protocol. This result is consistent across two independent experimental configurations (the original 200-epoch ablation run and this seeded, checkpoint-aligned rerun) and is not an artefact of training procedure or random initialisation. The finding has a clear mechanistic explanation: when two of three modalities carry no discriminative signal, a fixed gate that ignores them outperforms a learned gate that must discover this fact during training. The gate's benefit is therefore contingent

Table 10: Ablation study on OULAD. All variants trained under an identical protocol: up to 200 epochs with dual (AUC + F1) patience-20 early stopping, seed 42, mean AUC-ROC and Engagement-F1 over 5-fold CV. Best result in each column underlined; overall best in bold.

Variant	AUC ↑	Eng-F1 ↑	RMSE ↓
Behavioral only	$0.7336_{\pm 0.0082}$	0.4396	1.6222
Shared gate (one W all edges)	$0.8035_{\pm 0.0421}$	0.4721	1.5588
MG-HGNN full	$0.8346_{\pm 0.0352}$	0.4972	1.5427
Uniform gate ($\alpha = [\frac{1}{3}, \frac{1}{3}, \frac{1}{3}]$)	$0.8471_{\pm 0.0225}$	0.5087	1.5062
Struct only ($\alpha = [1, 0, 0]$)	$\mathbf{0.8873}_{\pm 0.0235}$	**0.5270**	1.5461

All five variants trained under the *same* epoch budget and early-stopping rule — this table specifically corrects an earlier version that compared 50-epoch ablation variants against a 200-epoch full model. Under a fair, identical protocol, the fixed struct-only gate outperforms the full learned gate by 0.053 AUC on OULAD; see discussion above.

on the presence of genuinely informative text content — a condition not met by any of the three currently evaluated datasets. Evaluation on MOOC-Cube, where real course descriptions and video titles are available, is identified as the necessary next step to validate the full three-modality gate mechanism. Preliminary evidence that the gate does not add substantial noise on text-free data (the AUC gap is 0.053, not catastrophic) is itself informative: it suggests the mechanism does not badly overfit to uninformative inputs, even though it does not match a fixed gate's performance here either.

Structured features are the strongest single modality on OULAD. The struct-only variant (0.8873 AUC at 200 epochs) outperforms behavioral-only (0.7336) by a wide margin, consistent with OULAD's rich demographic and academic background features. The full multi-modal model does *not* achieve the best RMSE at 200 epochs despite OULAD having no real text modality (text embeddings are placeholder zeros): Uniform gate's RMSE (1.5062) is lower than the full model's (1.5427). We note that on OULAD the gate does not suppress the placeholder text modality as initially hypothesised; see Section 5.5 for the full discussion.

5.7 Reproducibility and Runtime

All experiments were conducted on a MacBook Pro with an Apple M-series CPU (no GPU). Despite being a CPU-only configuration, the training pipeline is practical due to three design decisions. First, BERT embeddings are computed once and cached, eliminating the dominant transformer inference cost from the training loop. Second, the warmup-then-cache protocol for HGT and HAN baselines (3 full-graph epochs followed by 50 head-only epochs on frozen embeddings) avoids repeated full-graph forward passes. Third, the XGBoost baseline runs in a separate subprocess to avoid OpenMP conflicts with PyTorch's threading library on macOS.

Table 11 reports approximate wall-clock training times. MG-HGNN's longer per-fold training time (25 min) relative to the baselines reflects the larger model capacity (872K trainable parameters vs. HGT's ≈678K) and the 200-epoch budget with patience-based early stopping. On a GPU with torch-scatter installed, HGTConv's sparse message passing drops from $O(|\mathcal{E}| \cdot d^2/H)$ wall-clock to a scatter-reduce CUDA kernel, and we estimate the per-fold training

Table 11: Wall-clock training time per fold on CPU-only hardware. MG-HGNN times include BERT cache loading but exclude the one-time 90-second BERT inference precomputation.

Model	Time / Fold	Total (5-fold)
XGBoost (early feat.)	≈ 15 s	≈ 75 s
GAT (30 epochs)	≈ 4 min	≈ 20 min
HGT (warmup + heads)	≈ 8 min	≈ 40 min
HAN (warmup + heads)	≈ 6 min	≈ 30 min
MG-HGNN (200 ep, ES)	≈ 25 min	≈ 2 hr
BERT precomputation	≈ 90 s (once)	

time would fall to under 3 minutes based on the ratio of sparse-matmul vs. gather-scatter performance on comparable graphs reported by Hu et al. [5]. Code and pre-computed BERT embeddings are available at https://github.com/shiv3589/mg_hgnn.

6 Discussion

6.1 Why Relation-Conditioned Gating Works

The key empirical finding of this paper — that the modality gate contributes +0.047 AUC over the identical HGTConv backbone without gating on OULAD (Table 4) — demands an explanation grounded in the structure of OULAD. Per-task breakdown across all metrics is reported in the supplementary repository at https://github.com/shiv3589/mg_hgnn. This gap should be read alongside the ablation finding (Section 5.6) that a fixed struct-only gate beats the full learned gate by an even larger margin (0.053 AUC) on the same dataset — the learned gate helps relative to *no* gating mechanism (HGT), but a well-chosen *fixed* gate still does better than either on OULAD specifically.

The dataset contains four semantically distinct edge types connecting students to nodes of very different types (*course*, *peer*, *resource*, *assessment*). These node types carry fundamentally different feature distributions: course nodes have rich module descriptions (textual), resources are typed VLE activities (primarily structural metadata), and assessments carry score distributions (structured). When a student's gated embedding is constructed for the *enrolled_in* relation, what the course node needs to receive is: "who is this student, in terms of their academic and demographic background?" — a question answered almost entirely by the structured modality. When a resource node aggregates over its *accessed* neighbors, what matters is: "what kind of learner accesses me?" — a question answered by the sequential clickstream, not by demographic records.

Standard HGNN designs without gating force a single fused embedding to serve all of these roles simultaneously, creating a representational bottleneck. The modality gate removes this bottleneck with a minimal parameter cost (0.53%) by learning to route the appropriate signal to each relational context. The ablation result (shared gate: −0.031 AUC vs. full model) confirms that even a single globally learned mixture — which can adapt to the overall data distribution but not to individual relation types — fails to recover this performance.

6.2 OULAD and MOOC-Cube as Stress Tests for Text Modality

An important caveat of our evaluation is that OULAD contains no student-authored text: the student-level text field in our loader is a constant placeholder zero vector for every student, carrying no per-student discriminative signal by construction.

Contrary to our initial expectation, the gate does *not* reliably suppress this uninformative modality. Table 8 shows α_t ranging from 0.27 to 0.41 across relation types, and text receives the single *highest* weight of any modality for *accessed* edges ($\alpha_t = 0.41$ vs. $\alpha_s = 0.35$, $\alpha_b = 0.24$). Because the underlying embedding is a constant placeholder, this weight cannot reflect genuine text informativeness — it more plausibly reflects the gate's uninformed initialisation than a genuine assessment of text usefulness. We treat this as a genuine limitation of the current gate design on OULAD: α_t magnitude on this dataset should not be read as an interpretability signal for text usefulness.

This motivated evaluation on a dataset with rich text content, such as MOOC-Cube [28]. We have since run that follow-up (Section 5.4), and it does *not* resolve the question: MOOC-Cube's real text lives on course and video nodes, but the gate — in both this paper's original formulation and the evaluation just run — operates on *student*-level modality embeddings, and MOOC-Cube's per-student text field is exactly as uninformative a placeholder as OULAD's (Table 9). The originally proposed test was mis-scoped: swapping datasets was not sufficient, because the placeholder-text problem is a property of which node type the gate is applied to, not of which dataset supplies the surrounding graph. A genuine test requires extending gate computation to course- or resource-level embeddings, where real text actually enters the model on MOOC-Cube. We flag this as the necessary follow-up before treating gate weight magnitude as an interpretability signal in deployed settings.

6.3 Scalability Considerations

The primary scalability bottleneck of MG-HGNN in its current form is the HGTConv message passing, whose $O(|\mathcal{E}| \cdot d^2/H)$ per-layer cost grows with the total number of edges across all relation types. For OULAD with 652K edges this is manageable on CPU, but large-scale deployments (e.g., Coursera-scale MOOCs with millions of enrollment and access edges) would benefit from mini-batch training with neighborhood sampling. The modality gate itself scales perfectly: its $O(|\mathcal{E}| \cdot d)$ cost is strictly less than the backbone, and since the gate computation is a batched linear operation (one $\mathbb{R}^{3\times 3d}$ multiply per relation type), it adds no new bottleneck regardless of graph size.

An alternative training regime for large graphs is the warmup-then-cache protocol used for the HGT baseline: run 3–5 full-graph epochs to allow the gate to find a reasonable operating point, cache the gated student embeddings, then train prediction heads on the cached embeddings. This decouples the $O(|\mathcal{E}|)$ gate computation from the per-epoch training loop. However, in the full MG-HGNN training loop the gate is updated jointly with the backbone, allowing the gated embeddings to improve as the GNN backbone improves — a benefit that the cache protocol trades away. Future work should explore asynchronous cache refreshes every k epochs as a middle ground.

Empirical findings from scaling to MOOC-Cube. Attempting to train MG-HGNN on MOOC-Cube's 199,199-student graph surfaced two concrete scalability failure modes not visible at OULAD's scale (28,785 students). First, the full-graph, single-batch encoder used for OULAD triggered kernel OOM kills on a 32GB CPU instance even after edge subsampling (Section 5.1); this was the proximate reason for moving MOOC-Cube training to a GPU instance rather than a CPU one. Second, and more specifically, the bottleneck was not primarily the edge count but the behavioral encoder: a bidirectional 2-layer GRU invoked unbatched over all 199,199 student sequences allocates its raw output tensor at $N \times T \times d \approx 199{,}199 \times 50 \times 128$ elements (≈5.1GB in float32 alone, before GRU internal state), which we confirmed via `dmesg` and step-by-step RSS instrumentation was the actual crash point — not, as we first hypothesised, the graph construction step. Mini-batching this single call (batch size 8,192) resolved it without any change to model semantics. We record this because it generalises beyond MOOC-Cube: any dataset with student sequence counts in this range will hit the same GRU-memory wall regardless of edge count, and the fix is a few lines of batching, not an architecture change. We note this GPU migration changed only tensor placement, not the per-fold training schedule: as on OULAD, the graph encoder (including the gate) is still fit once per fold before the head-only epoch loop runs on cached embeddings (Section 4); moving to GPU resolved the memory ceiling but did not, by itself, enable the full end-to-end joint optimisation of encoders, gate, and heads that earlier drafts of this paper speculated GPU access would provide.

7 Conclusion

We presented MG-HGNN, a modality-gated heterogeneous graph neural network for student academic performance prediction. The central contribution is a lightweight, learnable modality gate that conditions multi-modal feature fusion on the edge relation type during message passing — the first such mechanism in the educational HGNN literature. The gate adds only 0.53% of trainable parameters and produces a relation-specific weighting whose learned coefficients can be inspected: structured features dominate *enrolled_in* and *collaborated_with*, while behavioural features lead for *submitted_to*, consistent with educational intuition. However, the gate does not cleanly suppress the placeholder text modality on *accessed* edges (Section 6), a limitation we discuss below.

Experiments on OULAD demonstrate that MG-HGNN is the only evaluated model that places in the top two across all three tasks — grade regression (RMSE 1.5453 ± 0.1170), dropout prediction (AUC-ROC 0.8443 ± 0.0292), and engagement classification (F1 0.5059 ± 0.0249). The critical comparison is with HGT — the identical backbone without gating — where the modality gate alone contributes +0.047 AUC (Table 4); per-task breakdown across all metrics is reported in the supplementary repository at https://github.com/shiv3589/mg_hgnn. This directly quantifies the value of relation-conditioned modal fusion. Ablation results confirm that edge-type conditioning is the load-bearing component: replacing per-relation gate matrices with a single shared matrix causes the largest single-component AUC drop (−0.031); notably, the uniform-gate variant achieves a *lower* (better) RMSE than the full learned gate (1.5062 vs. 1.5427), one of several signs that the learned gate's benefit on OULAD is not uniformly positive across metrics (Section 5.6).

Limitations. We have since evaluated MG-HGNN on MOOC-Cube (Section 5.4), which does carry real per-course text, reaching AUC-ROC 0.9713 ± 0.0096; however, this does not resolve the text-modality limitation as originally framed. The gate operates on *student*-level embeddings, and MOOC-Cube's per-student text field is a placeholder stub exactly like OULAD's — MOOC-Cube's real text lives on course and video nodes, outside the gate's current scope (Section 5.5). No baselines have yet been re-run on MOOC-Cube, so the AUC gap over OULAD should not be read as a clean win: MOOC-Cube's labels are proxies derived from the same watch-time signal that feeds the behavioral encoder, which plausibly inflates the gap through feature/label closeness rather than genuine multi-modal advantage. The current graph encoder is still updated only once per fold, before the head-only epoch loop; moving MOOC-Cube training to a GPU instance fixed a real memory ceiling (Section 6.3) but did not change this schedule, so full end-to-end joint optimisation of encoders, gate, and heads remains future work, not something GPU access alone provided. Gate weight differentiation also continues to show substantial fold-to-fold variance (std ≤ 0.15 on OULAD), and MOOC-Cube's *submitted_to* relation has zero edges (no problem/exercise entity exists in the official release), so that relation's gate weights are uninterpretable on this dataset. Finally, MOOC-Cube checkpoints are not tagged by seed or dataset, so only one seed's fold checkpoints survive sequential multi-seed runs — an infrastructure gap, not a modeling one, but one that constrained which checkpoint the gate-weight analysis in Section 5.5 could use.

Future work. Three directions are most promising. First, extending gate computation to course- or resource-level embeddings on MOOC-Cube, where real text actually enters the model, is now the concrete next step for testing whether the gate responds to genuine text informativeness — superseding the original framing, which assumed evaluating on a text-rich *dataset* was sufficient regardless of which node type the gate is applied to. Second, extending the gate to condition on both edge type *and* edge features (e.g., submission content) would push toward fully multi-modal relational reasoning. Third, applying the modality-gating principle to federated settings — where student data cannot be centralised across institutions — represents a high-impact research direction for privacy-preserving educational analytics. USTGA [37] demonstrates that explicit temporal adjacency across academic presentations improves robustness under feature noise; incorporating such temporal structure into MG-HGNN's heterogeneous graph is a natural direction for future work.

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